COMP 3105A — Assignment 2

Binary Classification: Primal, Adjoint and SVM Dual Formulations

Fall 2026

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Contribution statement: Both group members contributed equally (50% each) to completing Assignment 2.

Question 1(d) — Regularization Experiments

We evaluated primal squared hinge and hinge classifiers on the three synthetic generative models using 100 independent runs per setting, 100 training examples and 1,000 test examples per run. We tested λ = 0.001, 0.01, 0.1 and 1.0 using group seed 104. Each entry is the mean classification accuracy over 100 runs.

Table 1. Mean training accuracy

λ	Squared hinge			Hinge		
	Model 1	**Model 2**	**Model 3**	**Model 1**	**Model 2**	**Model 3**
0.001	99.99%	57.55%	87.36%	99.86%	61.53%	87.08%
0.01	99.91%	57.99%	87.17%	99.65%	62.47%	86.95%
0.1	99.71%	58.44%	87.29%	99.16%	62.01%	87.10%
1.0	99.24%	56.46%	87.34%	98.33%	60.52%	86.57%

Table 2. Mean test accuracy

λ	Squared hinge			Hinge		
	Model 1	**Model 2**	**Model 3**	**Model 1**	**Model 2**	**Model 3**
0.001	98.83%	54.68%	88.00%	98.80%	59.93%	87.83%
0.01	98.94%	54.04%	87.99%	98.71%	60.01%	87.76%
0.1	98.80%	55.44%	88.02%	98.31%	60.34%	87.82%
1.0	98.27%	54.19%	87.89%	97.45%	60.20%	87.19%

Question 1(e) — Findings

The linear primal models perform best on generative model 1: test accuracies are roughly 98–99% for both losses. This is expected because model 1 is separated by a linear decision boundary. Model 2 is nonlinear, so a straight-line classifier performs much worse, with squared hinge test accuracy around 54–55% and hinge test accuracy around 60%. For model 3, the primal linear classifiers reach approximately 88% test accuracy but cannot fully represent the curved structure of the two-moons data.

Across the four tested regularization strengths, accuracy changes are comparatively small and not consistently monotonic. Stronger regularization modestly reduces training accuracy on model 1, consistent with smaller model weights, but it does not systematically improve test accuracy. Hinge loss performs better than squared hinge on model 2 in this experiment, whereas their performances are similar on models 1 and 3.

Question 2(d) — Kernel Experiments

For the adjoint formulation, we fixed $\lambda = 0.001$ and evaluated five kernels across 10 independent runs. Each run used 100 training examples and 1,000 test examples for each generative model. The kernels were linear, polynomial with degrees 2 and 3, and Gaussian with widths 1.0 and 0.5. We used group seed 104. Mean accuracies over the 10 runs are reported below.

Table 3. Mean training accuracy

Kernel	Squared hinge			Hinge		
	Model 1	Model 2	Model 3	Model 1	Model 2	Model 3
Linear	100.00%	58.30%	87.30%	99.90%	62.10%	87.50%
Poly (d=2)	100.00%	100.00%	87.60%	100.00%	100.00%	87.90%
Poly (d=3)	100.00%	100.00%	100.00%	100.00%	100.00%	100.00%
Gaussian (σ=1.0)	100.00%	100.00%	100.00%	99.90%	100.00%	100.00%
Gaussian (σ=0.5)	100.00%	100.00%	100.00%	100.00%	99.90%	100.00%

Table 4. Mean test accuracy

Kernel	Squared hinge			Hinge		
	Model 1	Model 2	Model 3	Model 1	Model 2	Model 3
Linear	99.04%	55.66%	87.82%	98.85%	61.12%	87.56%
Poly (d=2)	97.83%	97.16%	87.89%	97.84%	97.09%	88.27%
Poly (d=3)	98.01%	96.27%	99.39%	97.98%	96.24%	99.43%
Gaussian (σ=1.0)	97.79%	97.38%	99.39%	97.65%	97.24%	99.33%
Gaussian (σ=0.5)	95.82%	94.99%	99.74%	95.85%	94.92%	99.74%

Question 2(e) — Findings

Kernel selection substantially changes generalization performance. The linear kernel is especially strong for model 1, with 99.04% squared-hinge and 98.85% hinge test accuracy. On model 2, the degree-2 polynomial and Gaussian kernels attain approximately 97% test accuracy, compared with 55.66% and 61.12% from the linear kernel. This supports the interpretation that model 2 has a nonlinear decision boundary that can be represented by a quadratic or other nonlinear feature map.

Model 3 benefits even more from nonlinear kernels: degree-3 polynomial and Gaussian kernels achieve about 99.3–99.7% test accuracy, whereas the linear kernel reaches about 88%. Gaussian width 0.5 gives the highest observed model-3 test accuracy, 99.74%, for both losses. Although several nonlinear models reach 100% training accuracy, their test scores vary. This demonstrates why training accuracy alone is not enough for selecting a model. The two losses give broadly similar results for a fixed kernel.

Question 3(c) — MNIST 4-versus-9 Cross-Validation

We trained a dual-form SVM on the provided handwritten-digit data. Digit 4 was mapped to class -1 and digit 9 to class $+1$. The 1,000 provided training images were normalized by dividing pixel values by 255. We used 5-fold cross-validation with shuffled folds and random seed 104 to compare two regularization strengths and three kernels. The same folds were reused for every combination.

Table 5. Mean 5-fold validation accuracy

λ	Linear	Polynomial (degree 2)	Gaussian (width 1.0)
0.001	95.30%	98.40%	48.20%
0.01	95.30%	98.40%	48.20%

Selected hyperparameters: λ = 0.001 and the degree-2 polynomial kernel, with average validation accuracy 98.40%. Both tested lambda values tie at 98.40% for degree 2; our selection procedure returns the first maximum and chooses 0.001.

The polynomial kernel outperformed the linear kernel (95.30%) and the tested Gaussian kernel (48.20%). It can capture nonlinear interactions among pixel features. These findings only compare the six tested combinations; they do not show that no other parameter combination could do better.

Question 3(d) — Available-data Verification

After selecting the hyperparameters, we retrained the dual SVM on all 1,000 available images and obtained 100.00% training accuracy. This is not a held-out test estimate. Accuracy on the instructor’s undisclosed test set cannot be reported because those data and labels are not available.

Implementation and Validation Notes

The algorithms were implemented directly using NumPy/SciPy and CVXOPT rather than prebuilt machine-learning estimators. Squared hinge objectives use SciPy minimize and hinge objectives use quadratic programming; the SVM dual is also a QP. For adjoint hinge, an equivalent SVM-dual numerical fallback handles cases where CVXOPT cannot confirm convergence for the larger adjoint QP. Local grader-style testing reported 10 passed, 0 failed; hidden instructor tests were not available.

Team Acknowledgment

Mohamed Azab (101331588): 50% contribution.
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